

AAA Concepts:

- o AAA is a term, which stands for Authentication, Authorization & Accounting.
- o AAA is a centralized management of users to access the network resources.
- o AAA services allow setting up access control on Cisco Routers or Cisco Switches.
- o Whenever user's attempts to login and access network it verifies by AAA database.
- o User management done on AAA database without need to reconfigure each device.
- o AAA also control connections passing through router for access network resources.
- o AAA provides flexibility and scalability, using privilege levels allows the flexibility.
- o AAA can be RADIUS Protocol or TACACS+ protocol where database is located.
- o Use local database fallback if the TACACS+/RADIUS server becomes unavailable.
- o Fallback only occurs if the AAA Server is unavailable or down for some reason.
- o There are two use cases of AAA, one is device administration other network access.

Authentication
Who are you?

Authorization
How much can you spend?

Accounting
What did you spend it on?

Statement of Personal Credit Card Account

Account Number: 1234-567-890 | Statement Closing Date: 01-31-01 | Current Amount Due: \$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99600-1234

MAIL PAYMENT TO:
THE BANK
132 VINE STREET
ANYTOWN, USA 67500-0010

872919345 00178255000000003

Detach here and return upper portion with check or money order. Do not staple or fold.

Statement of Personal Credit Card Account

Cardmember Name: JOE EMPLOYEE | Account Number: 1234-456-890 | Statement Closing Date: 01-31-01

Statement Date: 02-01-01 | Payment Due Date: 03-01-01

Closing Date: 01-31-01

Credit Limit: \$1,500.00 | Credit Available: \$1221.50

New Balance: \$278.50 | Minimum Payment Due: \$20.00

Account Summary

Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	\$278.50

Reference Number	Sold	Posted	Activity Since Last Statement	Amount
43210987	01-03	01-13	Payment, Thank You	-\$74.25
01234567	01-12	01-13	Wings 'N' Things Anytown, USA	\$25.25
78901234	01-14	01-17	Record Release Anytown, USA	\$40.00
45678901	01-14	01-17	Sports Stadium Anytown, USA	\$75.25
3210987	01-22	01-23	Tie Tack Anytown, USA	\$20.75
76543210	01-29	01-30	Electronic World Anytown, USA	\$89.25
23455678		01-30	Transaction Fees	\$3.00
34567890		01-01	Annual Fee	\$25.00

PAGE 1 OF 1

Authentication:

- o Check the user's credentials to verify their identity is authentication.
- o Authentication is the process of proving identity to the system.
- o Authentication is the validation of an identity or a credential.
- o The user or computer has to prove its identity to the server or client.
- o Log on to a PC with a user name and password you are authenticating.
- o Authentication does not determine what tasks the individual can do.
- o Authentication merely identifies and verifies who the person or system is.
- o Authentication is about who somebody is or who some device is.

Authorization:

- o Determines what the user is allowed to do is called Authorization.
- o Restrict the access to the user or the device is called Authorization.
- o Server determines client has permission to use a resource or access file.
- o Authorization is the process of verifying access to something.
- o Authorization is about what they are allowed to do inside the network.
- o Authorization is the function of specifying access rights to resources.
- o Authorization is the process to confirm what you are authorized to perform.

Implement Accounting:

- o In AAA model, accounting features is also very much important in security.
- o Accounting command tracking commands, services and resources used by user.
- o Accounting is the measure of resources consumed by a user during access.
- o Accounting, includes amount of time, amount of data user has send or received.
- o Accounting is carried in the form of logs of session statistics and usage information.
- o Accounting data is used for authorization control, resources utilization, billing & planning.
- o Accounting is also very much helpful to troubleshoot if network devices are not working.
- o Accounting is disabled by default in Authentication, Authorization & Accounting model.
- o AAA Accounting Types Network, Exec, Commands, Connection and Resource.
- o AAA basically, collects and stores information about logins and activity happen in network.
- o Recorded while accessing the specific device during specific time in the network.